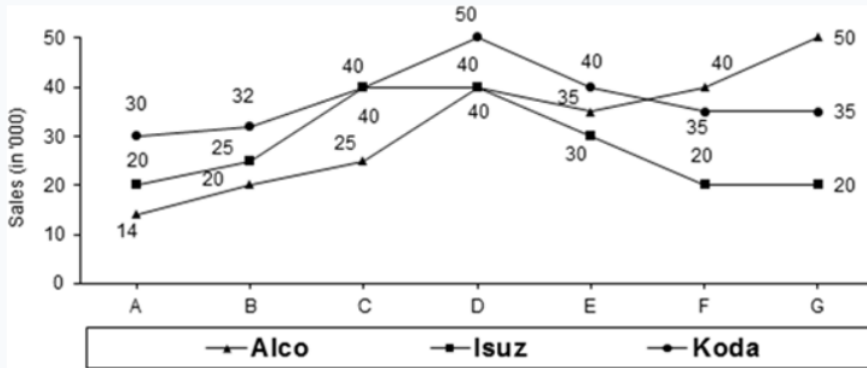


DI - LINE GRAPH

Directions for questions 1 to 5: Answer these questions on the basis of the information given below.

Study the following line graph which gives the details of the sales of three major car companies – Alco, Isuzu and koda – in seven states of a country in December 2018.



What is the approximate difference between the average sales of Koda and that of Alco for the given states?

☒ a) 5,430 (Your answer - ✓)

☐ b) 5,700

☐ c) 6,200

☐ d) 6,550

Solution:

Average sales of Koda

$$= \frac{30 + 32 + 40 + 50 + 40 + 35 + 35}{7} \times 1000$$

$$= \frac{262}{7} \times 1000 \approx 37,430$$

$$\text{Average sales of Alco} = \frac{224}{7} \times 1000 = 32,000$$

The required difference = 5,430.

Choice (A)

The sales of Koda in state E is approximately what percent of the total sales of the given cars in state E in the given month?

- ☐ a) 25%
- ☐ b) 38%
- ☐ c) 40%
- ☐ d) 30%

Solution:

Sales of Koda in E = 40,000

Total sales of cars in E = 1,05,000

$$\text{Required percentage} = \frac{40}{105} \times 100 = 38.09\%$$

Choice (B)

The sales of Alco in state D is what percent of the total sales of Alco in the given states?

- ☐ a) 35.45%
- ☐ b) 16.25%
- ☐ c) 18.5%
- ☐ d) 17.85%

Solution:

Sales of Alco in state D = 40,000.

Total sales of Alco = 2,24,000.

$$\text{The required percentage} = \frac{40}{224} \times 100 = 17.85\%$$

Choice (D)

If the sales of the three companies is 15% of the sales of all types of cars in the given states in December, then what is the approximate percentage contribution of the sales of Alco to the total sales of cars in the given states for the same month?

- ☐ a) 7%
- ☐ b) 6%
- ☐ c) 5%
- ☐ d) 3.5%

Solution:

The percentage sales of Alco cars of the three types of cars in December 2009

$$= \frac{224000}{224000 + 262000 + 195000} \times 100$$

$$= \frac{224000}{681000} \times 100 = 33\%$$

The percentage sales of Alco cars in the total

$$= 33 \times \frac{15}{100} = 5\%$$

Choice (C)

In December, what is the ratio of the total sales of Alco in B, E and F and that of Koda in A and C?

- ☐ a) 19 : 14
- ☐ b) 11 : 13
- ☐ c) 12 : 17
- ☐ d) 15 : 12

Solution:

The total sales of Alco in B, E and F = 95,000.

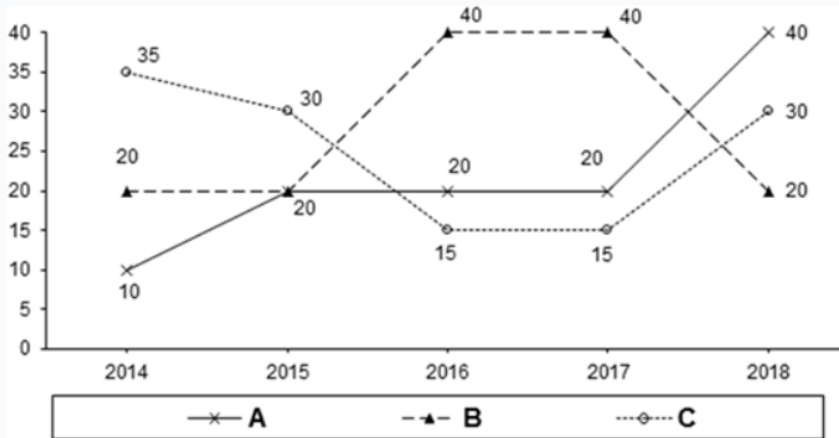
The total sales of Koda in A and C = 70,000.

The required ratio = 19 : 14.

Choice (A)

Directions for questions 6 to 10: Answer these questions on the basis of the information given below.

The line graph provides information about the percentage increase in the sales of the three products – A, B and C, as compared to previous year, across five years.



If the sales of A and B are in the ratio of 2 : 3 in 2014, what is the ratio of their sales in 2017?

☐ a) 42 : 49

☒ b) 24 : 49 (Correct Answer - ✓)

☐ c) 27 : 28

☐ d) 1 : 2

Solution:

Let the sales of A and B in 2014 be 200 and 300 respectively.

Sales of A in 2017 = $200 \times 1.2 \times 1.2 \times 1.2$
 $= 2 \times 172.8$

Sales of B in 2017 = $300 \times 1.2 \times 1.4 \times 1.4$
 $= 3 \times 1.2 \times 196$

The required ratio = 24 : 49

Choice (B)

Sales of product B in 2013 was Rs.12.5 million. Find the sales (in million) of B in 2016.

☐ a) 20.5

☐ b) 18.5

☐ c) 21.6

☐ d) 25.2

Solution:

$$\begin{aligned}\text{The sales of product B in 2016} &= 12.5 \times 1.2 \times 1.2 \times 1.4 \times 106 \\ &= 12.5 \times 2.016 \times 106 \\ &= 125 \times 2016 \times 102 \\ &= 25,200,000 \\ &= 25.2 \text{ millions}\end{aligned}$$

Choice (D)

If sales of A and B were equal in 2016, then what is was the ratio of their sales in 2018?

- ☐ a) 1 : 3
- ☐ b) 1 : 1
- ☐ c) 2 : 3
- ☐ d) 3 : 4

Solution:

$$\begin{aligned}\text{The required ratio} &= 1.2 \times 1.4 : 1.4 \times 1.2 \\ &= 1 : 1\end{aligned}$$

Choice (B)

Directions for questions 9 and 10: Type in your answer in the input box provided below the question.

For how many years from 2015 to 2018 (both inclusive), the change (in terms of percentage points) in the increase in sales of B was more than that of C?

Solution:

Year	Charge (in terms of percentage points)	
	B	C
2015	0	5
2016	20	15
2017	0	0
2018	20	15

It happened in two years 2016 and 2018.

Ans : (2)

Directions for questions 9 and 10: Type in your answer in the input box provided below the question.

If the sales of A was more than that of C in 2015, then in how many years (in the given period) was the sales of A more than that of C?

Solution:

By observation, the sales of A is more than that of C in 2016, 2017 and 2018.

Let sales of A in 2014 = a

Let sales of C in 2014 = c

So, sales of A in 2015 = 1.2a

Sales of C in 2015 = 1.3c

Given $1.2a > 1.3c$

$$\Rightarrow a > \frac{13}{12}c \Rightarrow a > c$$

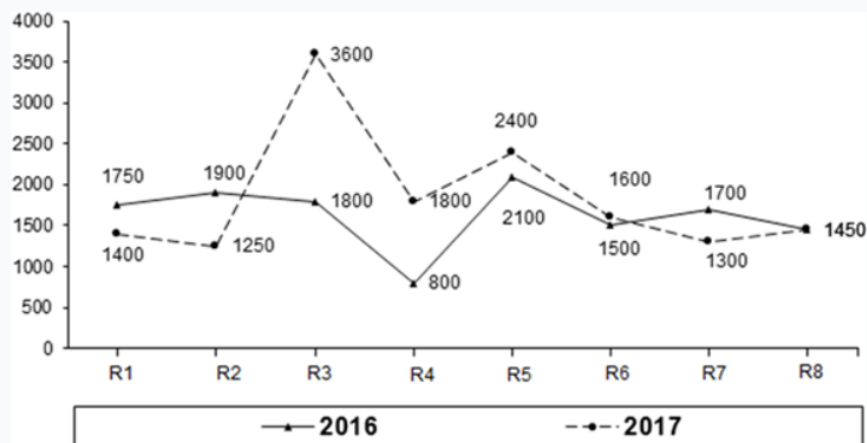
Hence, the sales of A is more than that of C in the year 2014.

$\therefore$ in all the five years, the sales of A is more than that of C.

Ans : (5)

Directions for questions 11 to 15: Answer these questions based on the information given below.

The graph gives information about the number of policies sold by a company in the different regions R_1 to R_8 of a country X, in the years 2016 and 2017.



The average number of policies sold by the company in all the given regions in 2016 was what percentage more/less than that in 2017?

- ☐ a) 12.16%
- ☐ b) 12.92%
- ☐ c) 13.57%
- ☐ d) 14.62%

Solution:

Region	2016	2017	Total
R ₁	1750	1400	3150
R ₂	1900	1250	3150
R ₃	1800	3600	5400
R ₄	800	1800	2600
R ₅	2100	2400	4500
R ₆	1500	1600	3100
R ₇	1700	1300	3000
R ₈	1450	1450	2900

The average number of policies sold in 2016 =
$$\frac{1750 + 1900 + 1700 + 1800 + 800 + 2100 + 1500 + 1450}{8}$$

= 1625

The average number of policies sold in 2017 =
$$\frac{1400 + 1250 + 3600 + 1800 + 2400 + 1600 + 1300 + 1450}{8}$$

= $\frac{14800}{8} = 1850$

∴ The required percentage = $\frac{1850 - 1625}{1850} \times 100$

≅ 12.16%

Choice (A)

If the total number of policies sold is the same in more than one region, then such regions are awarded the same rank. In which of the given regions is the total number of policies sold in both the years combined the 4th highest?

- ☐ a) R₁
- ☐ b) R₂
- ☐ c) R₆
- ☐ d) R₇

Solution:

Region	2016	2017	Total
R ₁	1750	1400	3150
R ₂	1900	1250	3150
R ₃	1800	3600	5400
R ₄	800	1800	2600
R ₅	2100	2400	4500
R ₆	1500	1600	3100
R ₇	1700	1300	3000
R ₈	1450	1450	2900

The region in which the number of policies sold is the 4th highest is R₆. Choice (C)

In which of the given regions was the percentage change in the number of policies sold from 2016 to 2017, the highest?

- ☐ a) R₃
- ☐ b) R₄
- ☐ c) R₈
- ☐ d) None of these

Solution:

Region	2016	2017	Total
R ₁	1750	1400	3150
R ₂	1900	1250	3150
R ₃	1800	3600	5400
R ₄	800	1800	2600
R ₅	2100	2400	4500
R ₆	1500	1600	3100
R ₇	1700	1300	3000
R ₈	1450	1450	2900

From the above table, the percentage change is the highest in region R₄ i.e.,
$$\frac{1800 - 800}{800} \times 100$$

= 125%

Choice (B)

Directions for questions 14 and 15: Type in your answer in the input box provided below the question.

What is the difference between the total number of policies sold in R_3 and R_4 combined and R_1 and R_5 combined in the given two years?

Solution:

$$\begin{aligned} &\text{The required difference} \\ &= (5400 + 2600) - (3150 + 4500) = 8000 - 7650 \\ &= 350 \end{aligned}$$

Ans : (350)

Directions for questions 14 and 15: Type in your answer in the input box provided below the question.

Find the total number of policies sold by the company in the two years in all the given regions.

Solution:

$$\begin{aligned} &\text{The total number of policies sold} \\ &= 3150 + 3150 + 5400 + 26000 + 4500 + 3100 + 3000 + 2900 = 27,800 \end{aligned}$$

Ans : (27,800)